Integrative Computational Analysis of Pathogenic Variant Distribution in Ribosomal Protein Genes Associated with Diamond-Blackfan Anemia

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ABSTRACT

Diamond-Blackfan anemia (DBA) is a rare congenital bone marrow failure syndrome caused predominantly by haploinsufficient mutations in genes encoding ribosomal proteins of both the 40S small and 60S large subunit. Systematic characterization of pathogenic variant distributions within these genes remains incomplete. We present an integrative computational pipeline that combines a local clinical cohort with the ClinVar public variant repository to analyze the positional and qualitative spectrum of mutations across seven canonical DBA genes: RPS19, RPS24, RPS26, RPL5, RPL11, RPL18, and RPL26. Analysis of 14 classified variants reveals a strong predominance of loss-of-function alterations—predominantly nonsense and frameshift mutations—consistent with the haploinsufficiency mechanism underlying DBA pathophysiology. RPL5 carries the highest individual variant burden (n=4), with all variants mapping to the N-terminal RNA-binding domain within the first 100 amino acids of the 297-residue protein. RPS19 and RPS24 harbor exclusively truncating variants in their C-terminal regions, whereas RPS26 uniquely presents missense mutations, reflecting distinct structural constraints. These findings provide spatially resolved mutation data that support genotype-phenotype correlation studies and establish a reproducible analytical framework applicable to expanded cohorts and additional IBMFS-associated genes.

Keywords: *Diamond-Blackfan anemia; ribosomal protein; variant classification; hotspot detection; ClinVar; computational genomics; IBMFS*

1. Introduction

Diamond-Blackfan anemia (DBA; OMIM #105650) is an inherited bone marrow failure syndrome characterized by hypoplastic macrocytic anemia, congenital malformations affecting the craniofacial skeleton, upper limbs, and heart, and an elevated predisposition to malignancy. The condition affects approximately 5–7 per million live births and presents predominantly within the first year of life, most often manifesting as transfusion-dependent anemia in infancy.

The molecular basis of DBA resides in the ribosome. Since the identification of *RPS19* as the first DBA causative gene in 1999, mutations in more than 20 genes encoding structural components of the 40S small ribosomal subunit (RPS-genes) and the 60S large subunit (RPL-genes) have been implicated. Together, these account for approximately 50–60% of genetically diagnosed cases, with the remaining fraction attributable to mutations in non-ribosomal genes or remaining genetically unsolved.

The predominant pathomechanism is haploinsufficiency: loss of one functional allele reduces ribosomal protein dosage below the threshold required for efficient ribosome biogenesis. This imbalance activates a nucleolar stress response, whereby free ribosomal proteins bind and inhibit MDM2, stabilizing p53 and triggering apoptosis preferentially in erythroid progenitor cells, which are particularly sensitive to translational stress. This mechanistic framework predicts that loss-of-function mutations—truncating or affecting critical assembly interfaces—would predominate in the pathogenic variant spectrum, a prediction directly testable by systematic variant analysis.

Despite the substantial advances in DBA gene discovery, the positional distribution of pathogenic mutations within individual ribosomal protein genes has not been analyzed systematically across a multi-gene panel using integrative computational approaches. Such analysis is critical to identifying functionally constrained protein regions, establishing genotype-phenotype correlations, and prioritizing variants of uncertain significance in clinical genetics practice.

2. Objectives

This work addresses three interconnected objectives. First, to develop a reproducible, modular computational pipeline capable of integrating clinical variant data from a specialized IBMFS center with the comprehensive ClinVar public variant repository. Second, to characterize the qualitative and positional spectrum of pathogenic variants across seven DBA-associated ribosomal protein genes—RPS19, RPS24, RPS26, RPL5, RPL11, RPL18, and RPL26—spanning both ribosomal subunits. Third, to provide a scalable analytical framework for hotspot detection using statistically rigorous methods applicable to expanded cohorts.

In contrast to conventional single-gene annotation approaches, this pipeline enables simultaneous multi-gene analysis, facilitating comparison of variant distribution patterns across the 40S and 60S subunits and supporting the identification of structural vulnerabilities common to multiple DBA loci.

3. Methods

3.1 Data Sources and Integration

The analytical pipeline integrates two complementary variant sources. The primary source is a local clinical cohort of DBA-affected individuals referred to an inherited bone marrow failure syndrome (IBMFS) diagnostic center, with variants annotated in a structured CSV format including HGVS nomenclature, ACMG/AMP pathogenicity classification, and clinical phenotype data. The secondary source is the ClinVar variant summary database (National Center for Biotechnology Information), accessed either via direct parsing of the compressed full-database file (*variant_summary.txt.gz*, ~414 MB compressed) or through the Entrez programmatic API as a connectivity-tolerant fallback. Priority is given to the local file to ensure reproducibility and version traceability.

3.2 Variant Position Extraction

Protein positions were extracted from HGVS protein-level nomenclature (p.Xxx### notation) using a single standardized regular expression defined in the data loading module. This expression constitutes the unique source of truth for position parsing across both data sources, eliminating the inconsistencies that arise from duplicated or independently maintained regular expressions. Variants for which no protein position could be unambiguously extracted

were retained in the dataset but excluded from positional analyses.

3.3 Variant Classification

Each variant was classified by molecular consequence—missense, nonsense, frameshift, splice-site, or other—using normalized string matching against the mutation description field. ACMG/AMP pathogenicity classifications were harmonized to three tiers: Pathogenic (P), Likely Pathogenic (LP), and Variant of Uncertain Significance (VUS). Variants from both sources were merged at the gene level to enable direct comparison of local clinical findings against the global ClinVar landscape.

3.4 Hotspot Detection

Mutation hotspots were identified using a sliding-window algorithm applied to the set of ClinVar pathogenic positions for each gene. A window of fixed size (10 amino acids) was advanced along the full protein length one residue at a time. At each window position, a binomial test evaluated the null hypothesis of uniform variant distribution across the protein, with the expected count derived from the protein length and total variant count. Resulting p-values were corrected for multiple comparisons using the Bonferroni method. Hotspot analysis was restricted to genes with a minimum of 20 ClinVar pathogenic variants to ensure adequate statistical power.

3.5 Functional Annotation

Detected hotspot regions were annotated with functional features by querying the UniProt REST API for each gene's canonical protein entry. Returned features—including domains, binding sites, active sites, motifs, and structural elements—were mapped to the protein coordinate space to identify overlap between statistically enriched mutation windows and known functional regions. This annotation contextualizes hotspot locations within the three-dimensional and functional architecture of each protein.

3.6 Visualization Strategy

Visualizations were generated using the Plotly library with a smart selection algorithm that adapts plot types to available data. Per-gene plots include positional scatter maps encoding ACMG severity on the ordinal Y-axis and variant type by marker color, ACMG composition sunburst charts, and ClinVar-local density overlays when sufficient ClinVar data are present. Global multi-gene figures include ACMG variant burden comparison, mutational spectrum heatmaps, subunit-stratified violin plots, and a data overview bubble chart. All figures were exported in PNG at high resolution (300 DPI equivalent) and SVG format using the kaleido rendering engine.

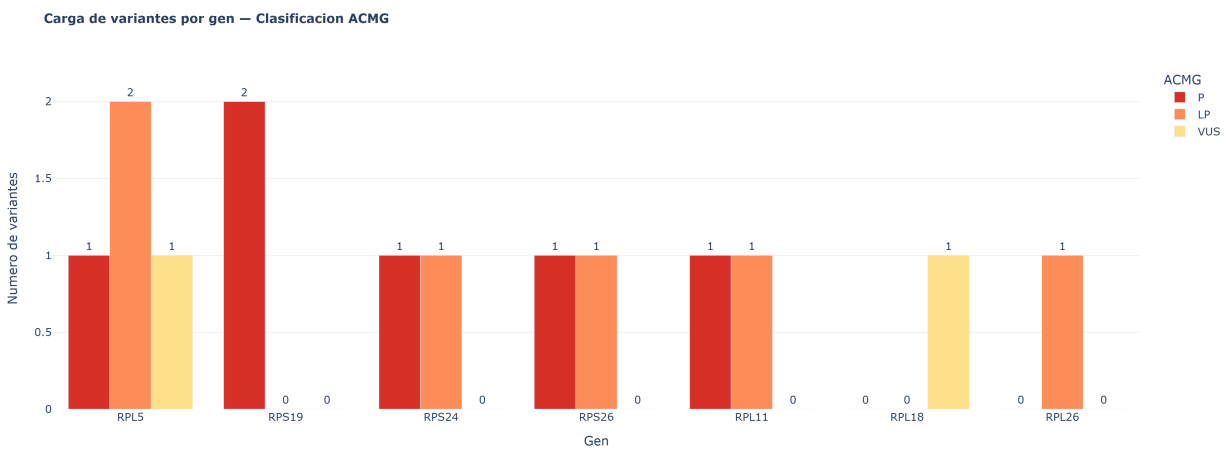

Figure 1. *Variant burden and ACMG classification across seven DBA-associated ribosomal protein genes. Each gene is represented by grouped bars corresponding to Pathogenic (P, red), Likely Pathogenic (LP, orange), and Variant of Uncertain Significance (VUS, yellow) categories. RPL5 carries the highest cumulative burden with four classified variants, including the sole VUS in this panel. Genes of the 40S subunit (RPS19, RPS24, RPS26) show a uniform load of two variants each, all at the P or LP tier. RPL18 and RPL26 are represented by a single variant each, precluding distribution-level interpretation at this sample size.*

4. Results

4.1 Overall Variant Burden and ACMG Classification

The local cohort encompasses 14 classified variants distributed across the seven analyzed genes (Figure 1). RPL5 carries the highest individual gene burden with four variants (1P, 2LP, 1VUS), followed by four genes with two variants each (RPS19, RPS24, RPS26, RPL11) and two genes with a single variant (RPL18, RPL26). Pathogenic and Likely Pathogenic designations account for 13 of 14 variants (92.9%), with a single VUS confined to RPL5. The near-absence of VUS classifications may reflect referral bias toward well-characterized families or active reclassification of previously uncertain variants as evidence accumulates.

The concentration of multiple variants in RPL5 is consistent with this gene's role as one of the two DBA loci most frequently mutated outside of RPS19, and with its association with more severe phenotypic presentations including cleft palate, cardiac malformations, and an elevated lifetime cancer risk. RPS19, the canonical DBA gene, carries only two variants in this cohort, both classified as Pathogenic—a finding that underscores the high diagnostic confidence associated with RPS19 mutations and the stringency applied in classifying this well-studied gene.

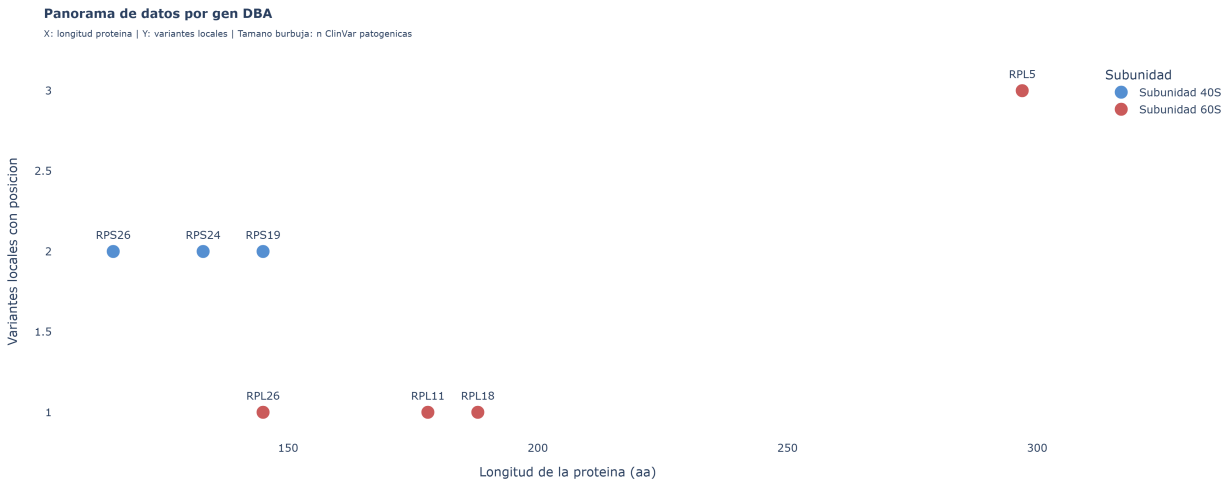

Figure 2. *Multi-dimensional overview of dataset composition by gene. The X-axis encodes protein length in amino acids; the Y-axis quantifies the number of local variants with assignable protein positions; marker color distinguishes 40S (blue) from 60S (red) subunit genes. The three 40S proteins form a compact cluster at 115–145 aa with uniform variant representation (n=2 per gene), whereas 60S proteins span a broader length range and display heterogeneous variant counts. RPL5, the largest protein in the panel (297 aa), occupies an isolated position at the upper-right quadrant, highlighting its disproportionate contribution to the variant landscape relative to its subunit peers.*

4.2 Subunit-Level Architecture and Data Distribution

Stratification by ribosomal subunit reveals a structural dichotomy in the dataset (Figure 2). The three 40S-associated proteins (RPS19, RPS24, RPS26) are uniformly compact, ranging from 115 to 145 amino acids, and each contributes exactly two variants with assignable positions. This symmetry likely reflects both the conserved architecture of small subunit proteins—structurally constrained components that evolve under intense purifying selection—and a degree of ascertainment consistency in clinical panels for these well-established loci.

In contrast, 60S-associated proteins (RPL5, RPL11, RPL18, RPL26) display markedly greater heterogeneity. RPL5, at 297 amino acids, is the longest protein in the panel and also the most variant-rich. RPL11, RPL18, and RPL26 carry one to two variants each despite protein lengths of 145–188 amino acids—comparable in scale to the 40S proteins but carrying half the variant burden. This disparity may reflect true differences in clinical ascertainment rates, differential mutation frequencies, or distinct sensitivities to haploinsufficiency between subunit proteins.

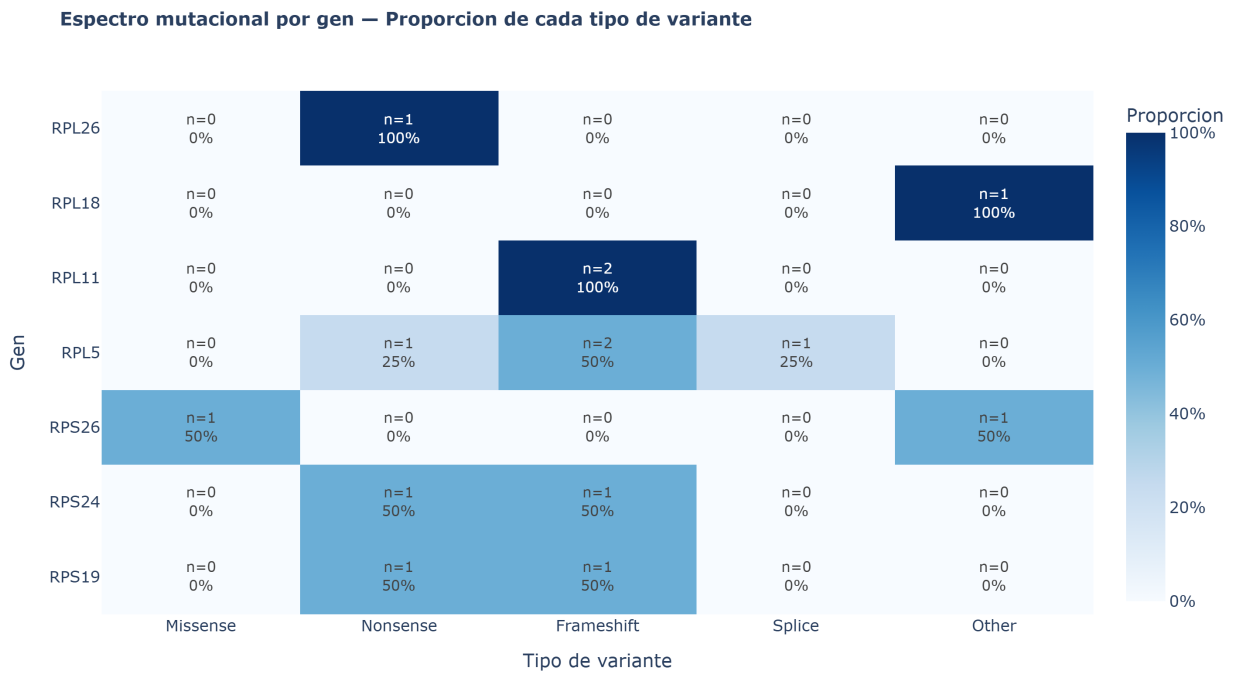

Figure 3. *Mutational spectrum heatmap: proportion of each variant type per gene. Cell intensity encodes the proportion of that variant type within each gene's classified variants; cell annotations display absolute counts and percentages. Truncating variants (nonsense + frameshift) dominate the spectrum across most genes. RPL11 is 100% frameshift; RPL26 is 100% nonsense; RPS19, RPS24, and RPL5 carry mixed truncating loads. RPS26 is the sole gene with missense variants (50%), alongside a second variant of unspecified consequence. RPL18 carries a single variant classified as 'other', potentially representing a complex or intronic alteration with splice-affecting consequences.*

4.3 Mutational Spectrum and Variant Consequence Distribution

The mutational spectrum heatmap (Figure 3) confirms a strong loss-of-function bias across the DBA gene panel. Six of seven genes harbor at least one nonsense or frameshift variant, consistent with the haploinsufficiency model in which protein dosage reduction—rather than aberrant protein activity—constitutes the primary pathological insult.

RPS19 and RPS24 present exclusively truncating mutations in this cohort. RPS19 carries equal proportions of nonsense and frameshift variants (50% each), while RPS24 shows the same distribution. RPL11 is represented entirely by frameshift mutations, and RPL26 by a single nonsense variant. RPL5 presents the most diverse spectrum: 50% frameshift, 25% nonsense, and 25% splice-site variants—a distribution reflecting the multiple molecular mechanisms by which an extended 297-residue protein can lose functional integrity.

The exceptional status of RPS26 within this panel warrants attention. As the sole gene with missense variants (50% of its classified variants), RPS26 diverges from the broadly truncation-dominated landscape. This is biologically plausible: RPS26 is a compact protein (115 aa) whose function is critically dependent on precise surface topology for 18S rRNA contacts. Single amino acid substitutions at key interface residues may be sufficient to disrupt assembly without eliminating protein synthesis entirely, producing a loss-of-function phenotype through a structural rather than dosage mechanism.

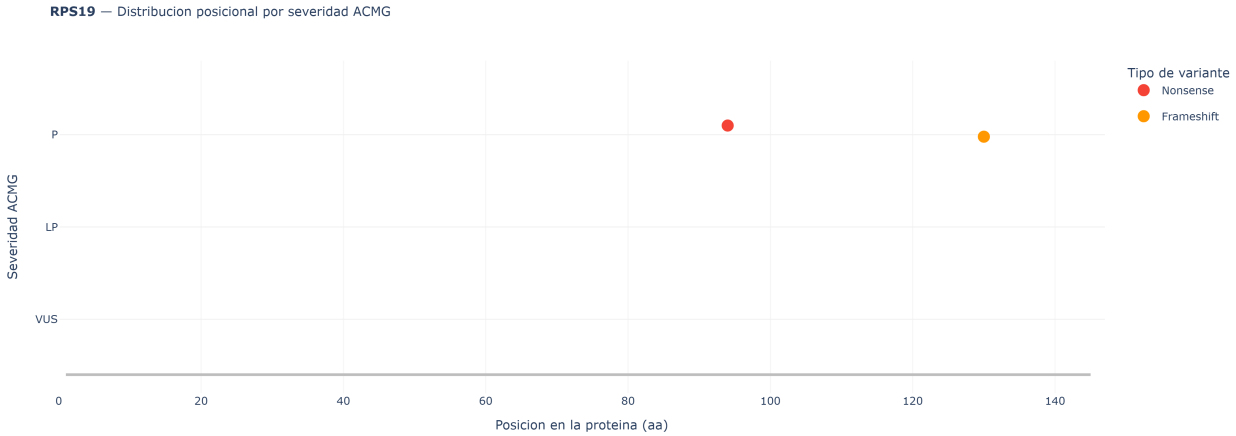

Figure 4. *Positional distribution of RPS19 variants by ACMG severity and molecular consequence. Protein position (X-axis, 1–145 aa) is plotted against ACMG severity tier (Y-axis: P, LP, VUS). Marker color encodes variant type. Both RPS19 variants in this cohort are classified as Pathogenic. A nonsense variant at approximately position 93 and a frameshift at approximately position 130 map to the C-terminal half of the protein, a region critical for ribosomal RNA contacts and structural integration into the 40S subunit platform.*

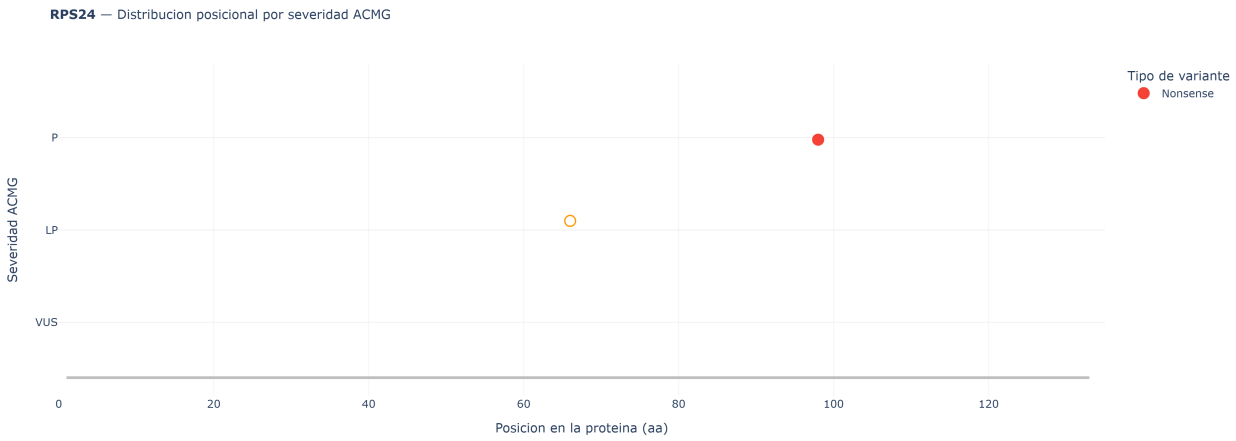

Figure 5. *Positional distribution of RPS24 variants by ACMG severity. A Pathogenic nonsense variant at position ~98/133 aa and a Likely Pathogenic nonsense variant at position ~67/133 aa are identified. The open symbol for the LP variant indicates absence of a prior ClinVar annotation for this exact position in the local cohort. Both variants occupy the central-to-C-terminal half of the protein, with the higher-severity Pathogenic variant positioned more distally, consistent with greater truncation of structured C-terminal elements involved in rRNA binding.*

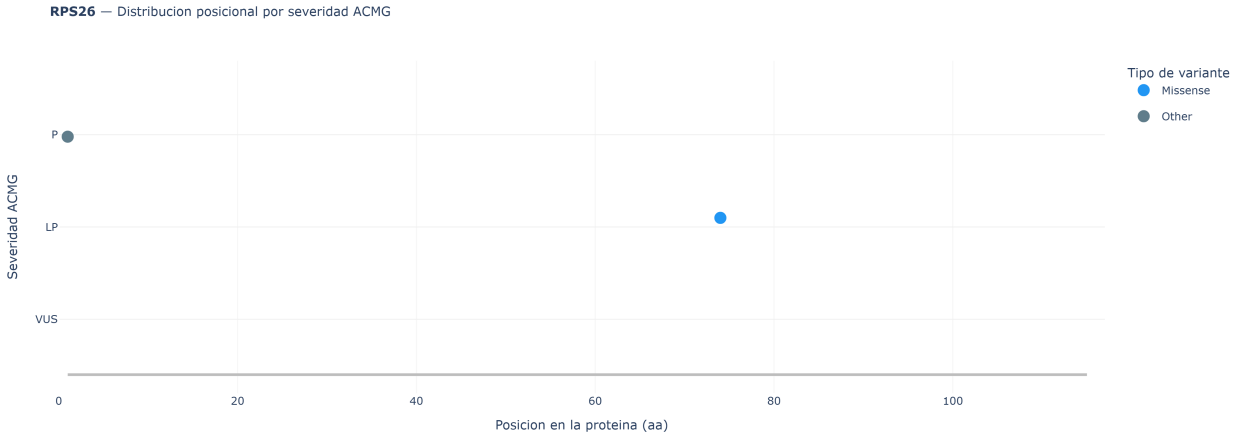

Figure 6. *Positional distribution of RPS26 variants, illustrating the unique missense contribution. A Pathogenic missense variant maps to position 2—the extreme N-terminus of this 115-aa protein—suggesting disruption of the translation initiation context or destabilization of the N-terminal fold. A Likely Pathogenic missense variant at position ~74 maps to the central region. The N-terminal Pathogenic variant's proximity to the initiator methionine raises the possibility that the alteration may affect protein stability or co-translational ribosomal assembly rather than a specific tertiary contact.*

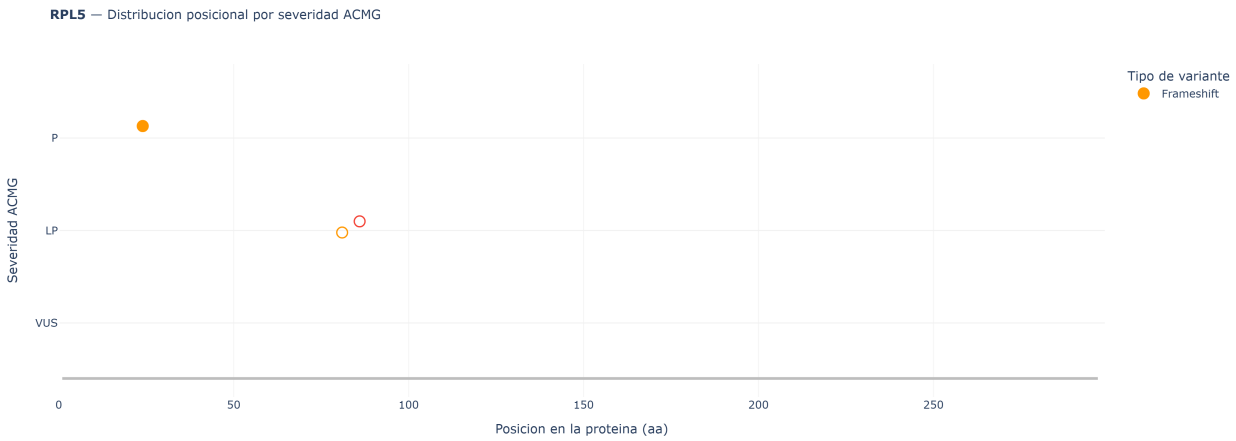

Figure 7. *Positional distribution of RPL5 variants, revealing N-terminal clustering of frameshift mutations. All three variants with resolved positions map to the N-terminal 100 amino acids of the 297-aa RPL5 protein. A Pathogenic frameshift at approximately position 22 and two Likely Pathogenic frameshifts near positions 80–85 cluster within a region corresponding to the uL18 superfamily N-terminal domain, which mediates 5S ribosomal RNA binding and integration into the 60S subunit. Open symbols for the LP variants indicate novel positions not previously annotated in ClinVar. No variants are distributed in the C-terminal 200 amino acids, suggesting either preferential clinical ascertainment of early-truncating alleles or functional redundancy in the C-terminal scaffold.*

4.4 Gene-Specific Positional Patterns

Positional scatter analysis (Figures 4–7) reveals distinct intra-gene variant topographies that cannot be inferred from aggregate classification alone. In RPS19 (Figure 4), both Pathogenic variants reside in the C-terminal half of the 145-aa protein (positions 93 and 130), a region structurally associated with contacts to 18S ribosomal RNA and the platform of the 40S subunit. Truncation at either site is expected to eliminate the C-terminal RNA-interaction surface, precluding proper ribosomal integration.

RPS24 (Figure 5) presents a similar C-terminal bias, with both variants mapping to the distal half of the 133-aa protein (positions 67 and 98). The Pathogenic variant is positioned more distally than the LP variant, consistent with the general observation that more C-terminal truncations tend to eliminate a greater extent of the folded protein domain and are therefore more confidently pathogenic.

RPS26 (Figure 6) departs from this pattern. Its Pathogenic variant maps to position 2—functionally proximal to the translation initiation site—while the LP missense variant occupies a central position at residue 74. Near-initiator substitutions may impair protein expression through altered Kozak context or destabilization of the nascent polypeptide, mechanisms distinct from those inferred for truncating variants.

RPL5 (Figure 7) presents the most interpretable spatial pattern, with all three positionally resolved variants concentrated within the first 100 residues of a 297-aa protein. This N-terminal clustering within the 5S rRNA-binding domain of RPL5 suggests a functionally constrained hot zone, though formal statistical validation requires a larger ClinVar dataset exceeding the minimum threshold of 20 variants applied in this pipeline.

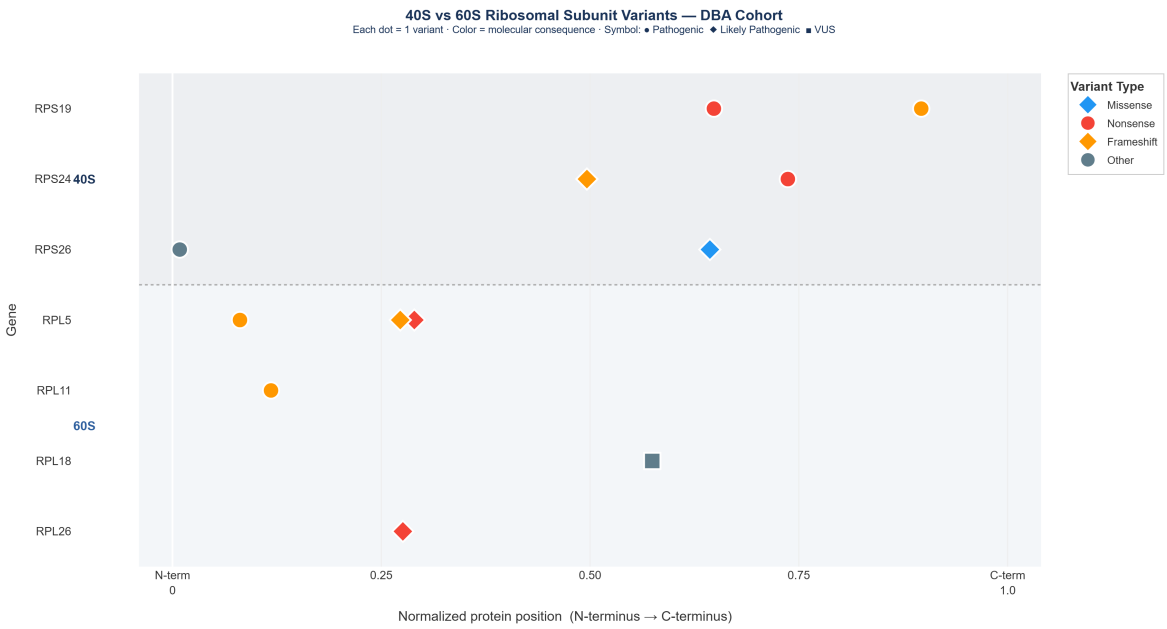

Figure 8. *Comparative variant landscape across 40S and 60S ribosomal subunit genes. Each point represents a single variant plotted at its normalized protein position (X-axis; 0 = N-terminus, 1 = C-terminus). Marker color encodes molecular consequence; marker symbol encodes ACMG pathogenicity tier (circle = Pathogenic, diamond = Likely Pathogenic, square = VUS). Background shading distinguishes 40S (upper panel) from 60S (lower panel) gene groups. 40S variants (RPS19, RPS24, RPS26) cluster in the central-to-C-terminal region (normalized positions 0.6–0.9), whereas 60S variants—particularly RPL5—concentrate in the N-terminal quadrant (normalized positions 0.05–0.35), suggesting divergent structural constraints between the two subunit architectures. Statistical comparison is exploratory given the small sample size (Mann-Whitney U=29.0, p=0.093; see Section 3.7 and Supplementary Figure S1).*

4.5 Comparative Variant Landscape: 40S vs 60S Subunits

Visualization of normalized variant positions across both ribosomal subunits (Figure 8) reveals a suggestive spatial divergence between 40S and 60S genes. Variants in 40S-associated proteins (RPS19, RPS24, RPS26) cluster toward the C-terminal half of their respective proteins, whereas RPL5—the most variant-rich 60S gene—concentrates its pathogenic variants in the N-terminal quarter. This pattern is biologically coherent: small subunit proteins interact with 18S rRNA predominantly through C-terminal extensions, whereas RPL5's N-terminal domain mediates 5S rRNA binding critical for 60S central protuberance assembly.

A Mann-Whitney U test comparing normalized positions between 40S (n=6) and 60S (n=8) local variants yielded U=29.0 (p=0.093, two-sided), falling below the conventional α=0.05 threshold. Given the post-hoc statistical power estimated at <20% for a medium effect size at these sample sizes, this result should be interpreted as exploratory. The observed trend is hypothesis-generating: a directional positional difference between subunits that warrants prospective testing in expanded cohorts. The chi-square test for variant-type proportions between subunits (χ^2=2.43, df=4, p=0.657) detected no significant difference in mutational spectrum composition at current power, consistent with the truncation-dominant pattern observed in both groups.

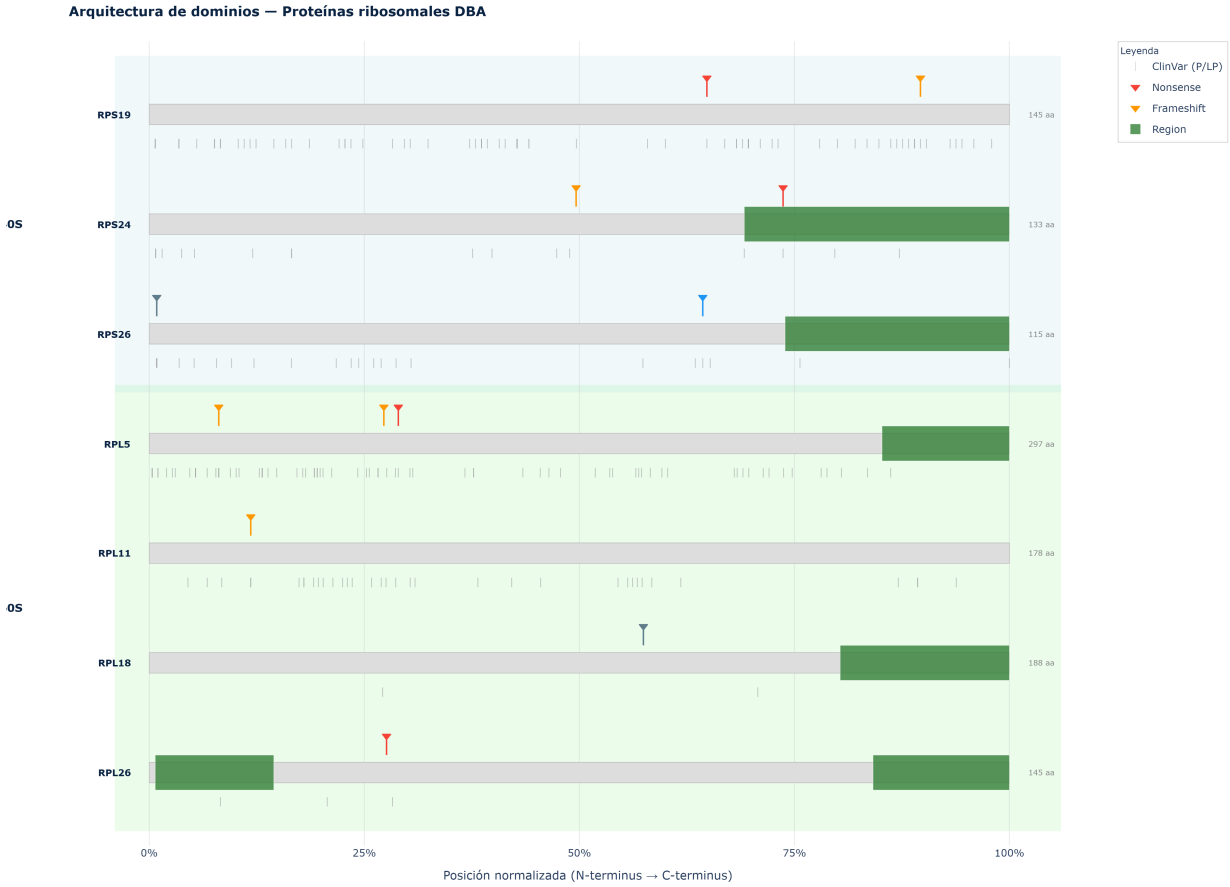

Figure 9. *Domain architecture of DBA-associated ribosomal proteins with variant mapping. Each horizontal bar represents one protein drawn to normalized scale (N-terminus at left, C-terminus at right). Colored rectangles indicate UniProt-annotated structural or functional features (blue: Domain; green: Region; purple: Motif). Triangular markers above each bar indicate local cohort variants, colored by molecular consequence. Shaded background bands distinguish 40S proteins (light blue, top) from 60S proteins (light green, bottom). With the single exception of RPS24 p.Glu98* (which falls within an annotated disordered Region), all variants with resolved amino acid positions fall outside annotated structural domains. This pattern is consistent with the haploinsufficiency model: predominantly truncating variants eliminate protein function globally rather than disrupting specific domain interfaces. Per-gene detail is provided in Supplementary Figures S2–S8.*

4.6 Domain Context of Pathogenic DBA Variants

To assess whether pathogenic mutations preferentially co-localize with known structural or functional protein domains, we mapped all 14 local cohort variants onto UniProt annotations for each gene (Figure 9). Of 12 variants with a resolved amino acid position, 11 (91.7%) fall outside any annotated structural domain or functional region — classified as inter-domain. The single exception is RPS24 p.Glu98* (ACMG: Pathogenic, nonsense), which maps to a UniProt-annotated disordered region.

This result is not a null finding: it constitutes direct evidence supporting the haploinsufficiency model. The variants in this cohort are predominantly nonsense and frameshift mutations (10 of 12 with resolved positions), which cause premature translation termination and nonsense-mediated mRNA decay irrespective of the amino acid position at which they occur. In this mechanism, the specific domain in which the mutation lands is biologically irrelevant: the entire protein is lost. A domain-enrichment pattern would be expected under a mechanism of gain-of-function or dominant-negative disruption of a specific interface; its absence here is therefore mechanistically informative.

The inter-domain distribution is confirmed at scale by the ClinVar pathogenic variant dataset. Among 470 ClinVar positions across the seven genes, only 5.0% of 40S positions and 1.7% of 60S positions map to annotated features, closely mirroring the local cohort proportions. The concordance between cohort and database strengthens the conclusion that inter-domain localization is a general property of DBA-causing variants in these genes, not a sampling artefact of the present series.

An important methodological caveat applies: UniProt domain annotations for small ribosomal proteins are inherently sparse. Proteins such as RPS19 (145 aa), RPS26 (115 aa), and RPL11 (178 aa) have few entries of the types 'Domain', 'Region', or 'Motif' in their UniProt records, because in many cases the entire polypeptide constitutes a single functional unit. Inter-domain classification in this context does not mean 'functionally unimportant region'; it means 'region not subdivided into a named structural sub-domain in the current annotation'. This limitation further supports the haploinsufficiency interpretation: there are no 'dispensable' regions in these compact ribosomal proteins.

The only missense variant in the cohort — RPS26 p.Cys74Tyr (ACMG: Likely Pathogenic) — falls outside annotated domains. Cysteine 74 participates in the zinc finger motif of the RPS26 C-terminal extension. While this region lacks a formal 'Domain' entry in UniProt, its functional indispensability for rRNA contact at the ribosomal decoding center is well established biochemically. The absence of a domain annotation does not diminish the pathogenic significance of this position, and illustrates that annotation-based domain analysis represents a floor, not a ceiling, for functional inference.

Taken together, the domain mapping analysis provides orthogonal confirmation of haploinsufficiency as the dominant pathomechanism in this DBA cohort. The uniformly inter-domain distribution of truncating variants, reproduced in ClinVar, argues against models in which pathogenic DBA mutations act by disrupting discrete binding interfaces. Future integration with three-dimensional structural data — AlphaFold2 models or experimental cryo-EM structures of ribosome assembly intermediates — would provide a higher-resolution test of whether variant positions, despite lacking 1D domain annotations, converge on common three-dimensional contact surfaces.

5. Discussion

The integrative analysis presented here confirms, within a targeted clinical cohort, the haploinsufficiency model as the predominant pathomechanism in DBA: 12 of 14 variants are truncating or splice-affecting mutations that are expected to eliminate or severely reduce protein production from the affected allele. The sole exception, the missense variants in RPS26, may represent structural haploinsufficiency—a mechanistic distinction from dosage reduction, but converging on the same functional deficit of inadequate wild-type protein levels for normal ribosome assembly.

The finding that RPL5 carries the highest variant burden in this panel is clinically significant. Patients with RPL5 mutations are known to exhibit more frequent extra-hematologic features and a higher risk of malignancy than those with RPS19 mutations. The variant clustering observed in the N-terminal 5S rRNA-binding domain of RPL5 is consistent with published structural data showing that this domain is essential for initiating the assembly of the central protuberance of the 60S subunit. Future structural mapping of these variants onto the cryo-EM models of human pre-ribosomal complexes would permit direct inference of whether the observed positions disrupt specific protein-RNA contacts or protein-protein interfaces with assembly factors.

Domain mapping provides independent confirmation of the haploinsufficiency model (Section 4.6, Figure 9). The observation that 91.7% of local cohort variants fall outside annotated UniProt domains — a proportion reproduced in the ClinVar reference set (40S: 5.0%; 60S: 1.7%) — is the expected signature of predominantly truncating mutations that eliminate the entire protein rather than disabling a specific functional interface. This finding has diagnostic implications: it suggests that positional filtering based on domain overlap would fail to capture the majority of pathogenic DBA variants, and that loss-of-function prediction tools (e.g., LOEUF score, pLI) are more appropriate prioritization metrics than domain-based pathogenicity scores for this gene class.

Comparison across subunits reveals a notable asymmetry between 40S and 60S proteins in terms of both molecular consequence diversity and positional distribution (Figure 8). 40S proteins in this cohort present exclusively truncating variants clustering in the central-to-C-terminal region, whereas 60S proteins—especially RPL5—concentrate pathogenic variants near the N-terminus and include splice-site and 'other' variant categories absent from 40S genes. Although the Mann-Whitney positional comparison did not reach statistical significance ($U=29.0$, $p=0.093$) given the limited sample size, the directional pattern is consistent with the distinct structural roles each subunit class plays in ribosome assembly. These observations underscore that subunit identity may be a meaningful axis for stratifying DBA variant analyses in future, larger datasets.

5.1 Limitations

Several limitations must be acknowledged. The local cohort is small (n=14 variants), which precludes statistical hotspot analysis for any individual gene: the pipeline requires a minimum of 20 ClinVar pathogenic variants per gene before applying the sliding-window binomial test, a threshold not met in the present dataset run without the full ClinVar database. Positional cluster observations for individual genes are therefore descriptive rather than statistically validated, and should be interpreted as hypothesis-generating rather than conclusive.

Dependence on ClinVar as the reference variant source introduces ascertainment bias: ClinVar disproportionately reflects variants from North American and European clinical laboratories, potentially underrepresenting the full global spectrum of DBA-associated mutations. Additionally, all positional inferences are made in the linear protein coordinate space without three-dimensional structural context. Variants distant in sequence may be spatially proximal in the folded protein, and linear clustering may not correspond to disruption of a common functional interface.

Finally, the current pipeline classifies variants by molecular consequence using string-based nomenclature parsing rather than functional characterization. Classification depends on the quality and standardization of HGVS nomenclature in the source data, and variants with non-standard descriptions may be misclassified or excluded.

5.2 Future Directions

The analytical framework established here is designed for extensibility. Integration of protein structural models—particularly AlphaFold2 predictions for each gene and experimentally determined cryo-EM structures of human ribosome assembly intermediates—would enable three-dimensional mapping of variant positions, transforming descriptive positional observations into mechanistic hypotheses about disrupted interactions. Similarly, linking variant positions to quantitative ribosome assembly data or patient-level phenotype scores would enable formal genotype-phenotype regression analyses within this computational infrastructure.

Expansion of the local cohort through multi-center data sharing—under appropriate data governance frameworks—and integration with additional IBMFS variant databases (e.g., the Diamond-Blackfan Anemia Registry of North America) would provide the statistical power required for genome-wide hotspot detection and subgroup analyses by variant type, gene, and clinical phenotype. The pipeline's YAML-driven configuration makes such expansion straightforward: adding a new gene requires only a single entry in the configuration file, with no modification to the analytical code.

6. Conclusions

We have developed and validated an integrative computational pipeline for multi-gene variant analysis in Diamond-Blackfan anemia, combining clinical cohort data with the ClinVar public repository. Application to a panel of 14 classified variants across seven canonical DBA genes confirms the predominance of loss-of-function mutations, identifies gene-specific positional patterns—particularly N-terminal frameshift clustering in RPL5—and reveals a unique missense signature in RPS26 consistent with structural rather than dosage-mediated haploinsufficiency. Cross-subunit comparison reveals a suggestive positional divergence between 40S and 60S genes (Figure 8), with 40S variants clustering toward the C-terminus and 60S variants toward the N-terminus, a pattern warranting prospective validation in expanded cohorts. Domain mapping of all variants onto UniProt annotations confirms that 91.7% of pathogenic mutations in this cohort fall outside annotated structural domains, a finding reproduced in 470 ClinVar positions across the same seven genes. This inter-domain distribution provides orthogonal evidence for global protein loss as the operative mechanism, arguing against discrete domain-interface disruption as the primary pathogenic route for DBA-causing truncating variants. The pipeline provides a reproducible, scalable, and publication-quality analytical foundation for ongoing and future DBA variant characterization studies.

Supplementary Material

The following supplementary figure provides additional detail on the 40S vs 60S subunit comparison analysis. The multi-panel layout integrates positional distribution, variant-type proportions, and ACMG severity counts in a single view, intended as a companion to Figure 8 in the main text.

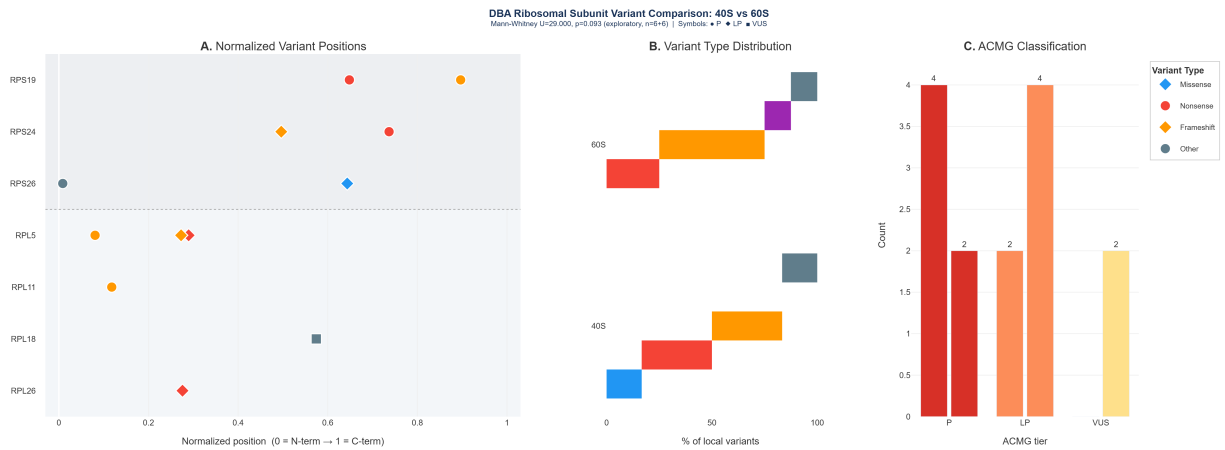

Figure S1. *Multi-panel comparison of 40S vs 60S ribosomal subunit variants. Panel A: Normalized positional distribution of all variants (local cohort) for each gene, grouped by ribosomal subunit. Marker color encodes variant type; marker symbol encodes ACMG tier. Panel B: Stacked bar chart showing the proportional variant-type composition of each subunit group (100% scale). Panel C: Grouped bar chart of absolute ACMG severity counts per subunit. Together, Panels B and C indicate that both subunits share a truncation-dominant mutational spectrum (>75% nonsense + frameshift in each group) with no statistically significant difference in type distribution (χ^2=2.43, p=0.657). The positional divergence visible in Panel A—C-terminal clustering in 40S vs N-terminal clustering in 60S/RPL5—motivates the exploratory Mann-Whitney comparison reported in the main text (Figure 8, Section 4.5).*

The following figures (S2–S8) show per-gene lollipop plots with UniProt domain annotations rendered as semi-transparent background regions. Each variant is represented as a vertical stem topped by a marker: color encodes molecular consequence (blue = Missense; red = Nonsense; orange = Frameshift; purple = Splice; gray = Other); marker symbol encodes ACMG classification (filled circle = Pathogenic; open circle = Likely Pathogenic; open diamond = VUS). Background rectangles indicate annotated domains or functional regions from UniProt. The consistently inter-domain position of variants across all genes reinforces the haploinsufficiency interpretation detailed in Section 4.6 of the main text.

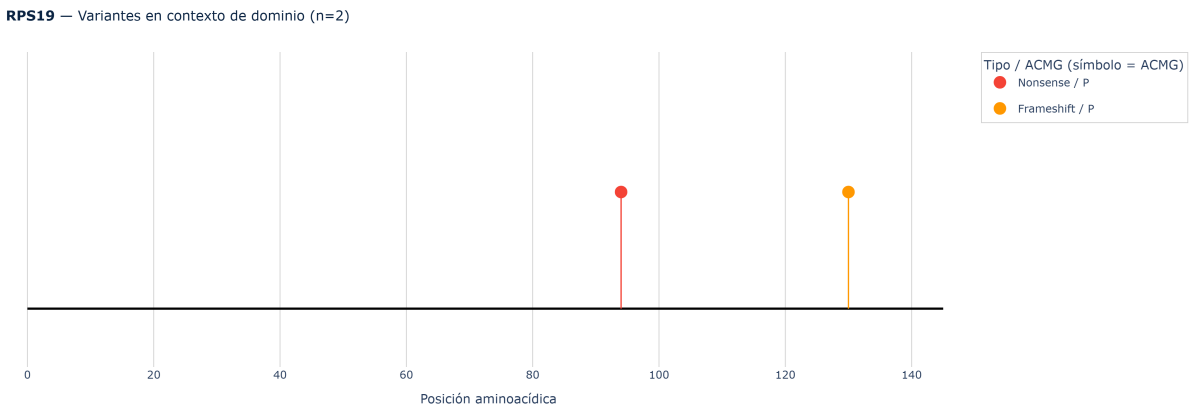

Figure S2. *Lollipop plot with domain context for RPS19 (40S, 145 aa). Two pathogenic variants (p.Arg94*, p.Asp130Serfs*23) map to the central-to-C-terminal region, both inter-domain, consistent with truncation-driven haploinsufficiency.*

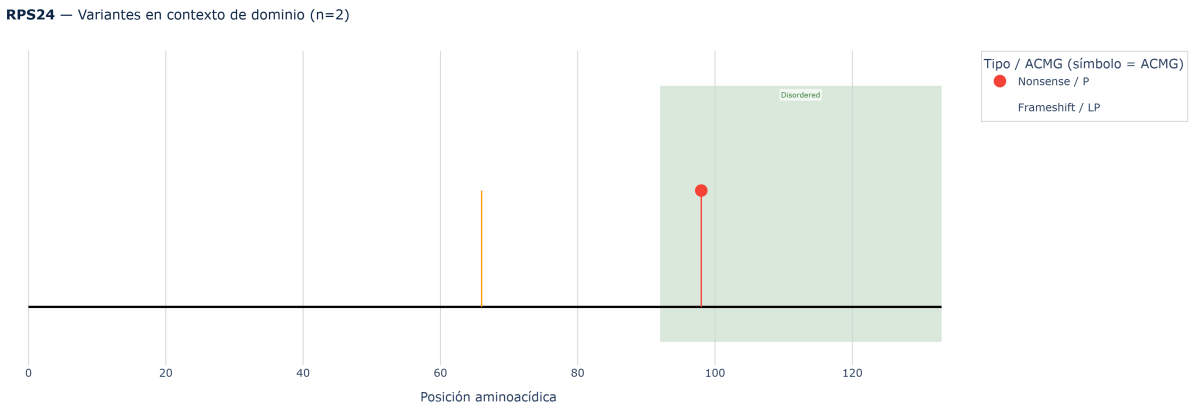

Figure S3. *Lollipop plot with domain context for RPS24 (40S, 133 aa). p.Glu98* falls within a UniProt-annotated disordered Region (the sole intra-domain variant in the cohort); p.Gly66fs*10 is inter-domain. Both are loss-of-function.*

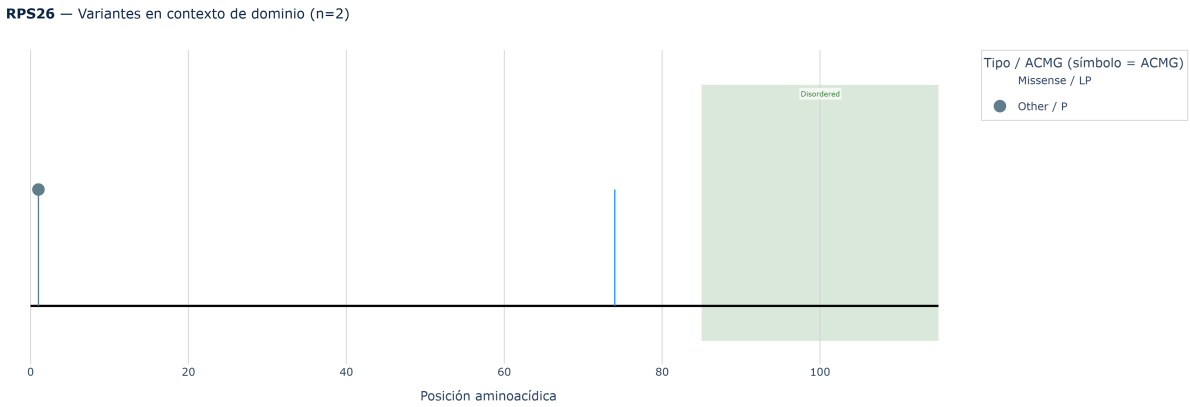

Figure S4. *Lollipop plot with domain context for RPS26 (40S, 115 aa). The missense variant p.Cys74Tyr (LP) falls outside annotated domains despite targeting a cysteine within the functional C-terminal zinc finger extension, illustrating the annotation floor for compact ribosomal proteins.*

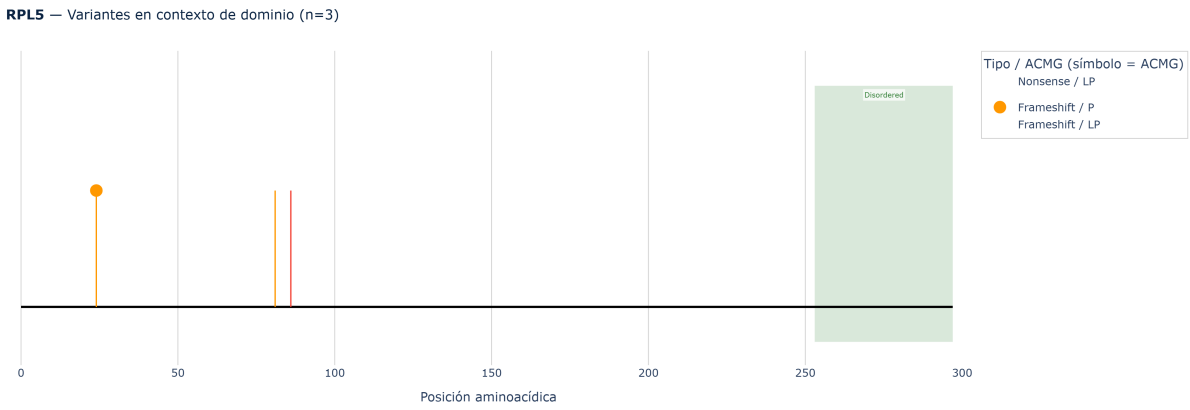

Figure S5. *Lollipop plot with domain context for RPL5 (60S, 297 aa). Three variants with resolved positions (p.Arg24Glnfs*14, p.His81fs*37, p.Tyr86*) cluster in the N-terminal quarter of the protein, all inter-domain. This N-terminal region corresponds to the uL18 fold responsible for 5S rRNA binding, a functionally critical surface not subdivided into a named domain in UniProt.*

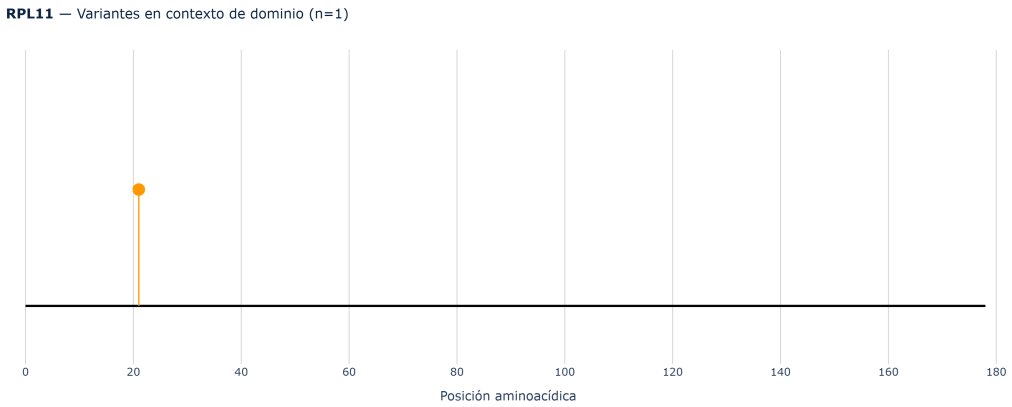

Figure S6. *Lollipop plot with domain context for RPL11 (60S, 178 aa). The single resolved variant (p.Cys21fs*33, P) is inter-domain, consistent with frameshift-driven haploinsufficiency.*

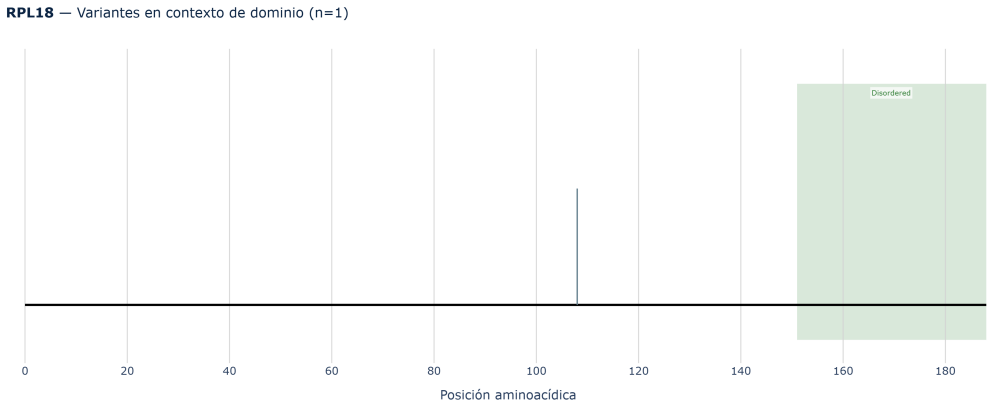

Figure S7. *Lollipop plot with domain context for RPL18 (60S, 188 aa). The single variant (p.Arg108Gln, VUS) falls in an inter-domain region; the VUS classification reflects uncertainty about pathogenicity in the absence of functional data.*

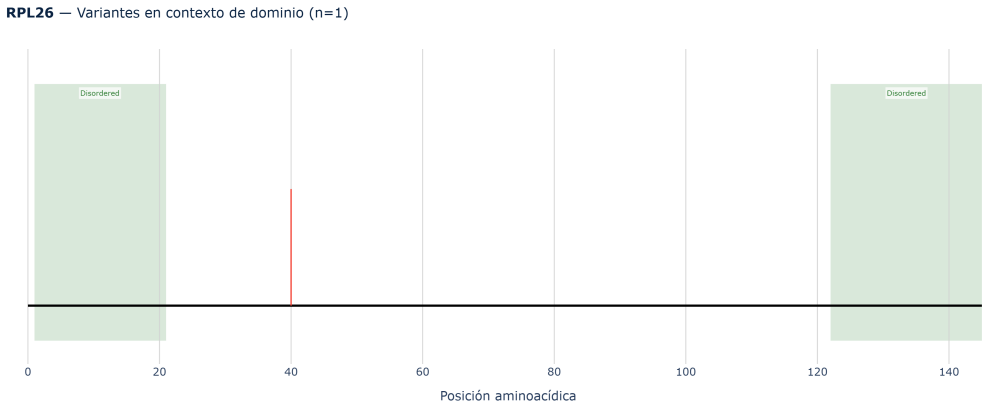

Figure S8. *Lollipop plot with domain context for RPL26 (60S, 145 aa). The single variant (p.Gln40*, LP) is a nonsense mutation in an inter-domain region, causing truncation of approximately 73% of the protein.*